

PROJECT COMPLETION DOCUMENT

Final Deliverables Documentation

FCAI E-Club Centralized Web Platform Project

PROJECT NAME

FCAI E-Club Centralized Platform

CLIENT NAME

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CONTACT & DATE

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1. PROJECT OVERVIEW

Summary: The FCAI E-Club platform is a comprehensive, centralized full-stack digital hub designed to connect academic theory with industry practice for engineering and entrepreneurial students at FCAI-CU.

Objective: To establish brand authority, streamline member management, automate event and workshop registrations, collect structured applicant CVs, and enable seamless corporate partner collaboration.

Scope: Included complete UX/UI design system implementation, Next.js 16 full-stack web application, Supabase database schema and RLS security policies, multi-step workshop registration pipeline, E2E test suites, and operational documentation.

2. FINAL DELIVERABLES SUMMARY

Deliverable	Type	Description & File Reference
Full-Stack Application	Codebase	Next.js 16, TypeScript, Tailwind CSS 4 setup with App Router (`/`, `/about`, `/team`, `/partnerships`, `/events`).
Database & Security	Database	PostgreSQL schema with 10 tables, auto-profile trigger, and 41 RLS security policies (`schema.sql`, `rls.sql`, `seed.sql`).
Server Actions & Storage	Backend	Server Actions for auth, proposal submission, multi-step registrations, and secure CV storage bucket pipeline.
Documentation Suite	Docs	Full set of setup guides, RLS reference, API/Server Action reference, maintenance playbooks, and testing guides.
E2E Test Suites	Testing	Playwright automated tests for registration pipelines, input validation, and CV upload constraints (`e2e/`).

3. EXPLANATIONS & SETUP INSTRUCTIONS

A. Environment & Local Server Setup

Run `pnpm install` to install all dependencies. Configure `.env.local` with `NEXT_PUBLIC_SUPABASE_URL` and `NEXT_PUBLIC_SUPABASE_ANON_KEY`. Launch local development with `pnpm dev` (accessible at `http://localhost:3000`).

B. Database & Storage Initialization

Execute `database/schema.sql`, `database/rls.sql`, and `database/seed.sql` in the Supabase SQL Editor. Ensure storage buckets `avatars` (public), `event-images` (public), and `cv-uploads` (private) are provisioned.

C. Analytics & Monitoring

Vercel Analytics is integrated for Core Web Vitals tracking. Sentry monitoring files (`sentry.client.config.ts`, `sentry.server.config.ts`) are pre-configured; set `NEXT_PUBLIC_SENTRY_DSN` in production environment variables.

4. ADDITIONAL NOTES

- **Postponement of Auth (Phase 4):** Direct auth enforcement on event registration was made optional as per project roadmap adjustment, allowing public submissions while preserving background email rate-limiting.
- **File Storage Security:** CV uploads are stored in a private bucket accessible only to authorized administrators through secure path metadata.
- **Design Consistency:** All custom UI components utilize the standardized `ds-*` utility class layer configured in `app/globals.css`.

5. FEEDBACK & REVISIONS HISTORY

Version	Feedback / Request	Resolution / Action Taken
v1.0	Add custom screening questionnaire & CV screening step to event flow.	Implemented 3-step registration form with file upload validation and server action processing.
v1.1	Include system maintenance & testing instructions for handoff.	Authored <code>MAINTENANCE-PLAYBOOK.md</code> and <code>E2E-TESTING.md</code> guides.

6. CLOSING REMARKS

This completion documentation marks the successful final delivery of the FCAI E-Club platform codebase and infrastructure. The system is fully equipped to scale event operations, manage community growth, and showcase student engineering impact. All codebase materials, database migration scripts, test suites, and operational guides have been handed over for ongoing management.

